

Business Summary: Credit Risk Decision Strategy

This report translates model probabilities into approve, manual review and decline decisions. It summarises portfolio mix, observed default performance, expected loss and threshold trade-offs.

Executive Summary

The validation portfolio contains **61,503** applicants with an actual default rate of **8.07%**. The current strategy approves **9.90%**, routes **38.02%** to manual review and declines **52.08%**.

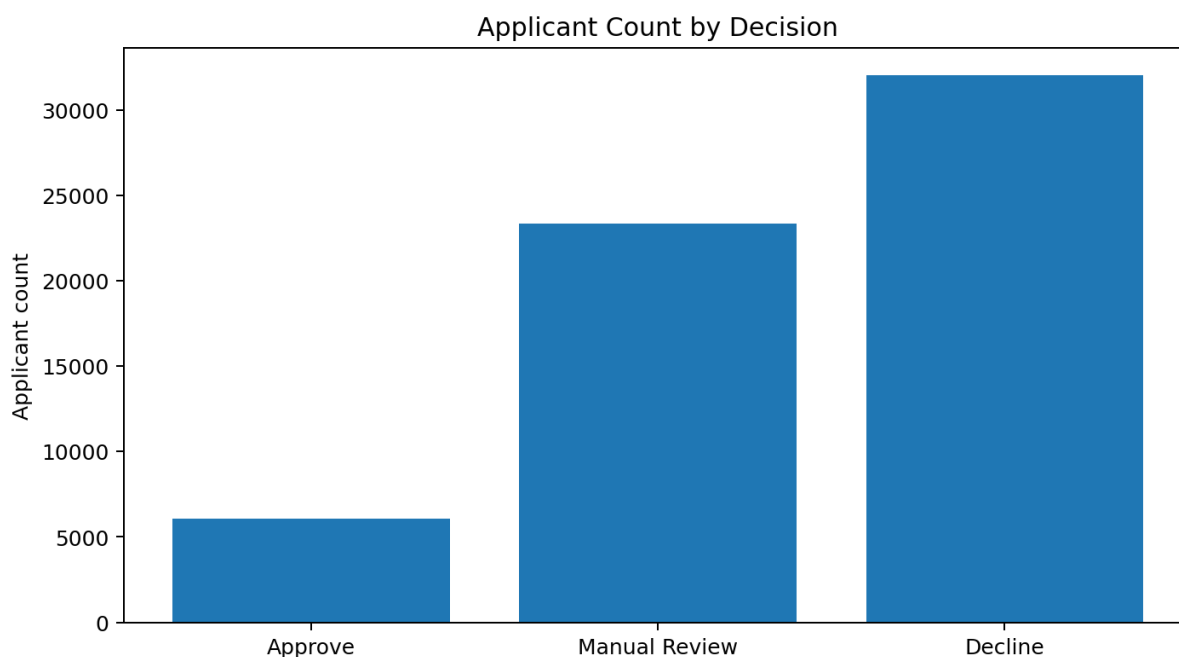
Decision Framework

Probability of Default	Risk Band	Decision
< 10%	Low Risk	Approve
10% to 30%	Medium Risk	Manual Review
> 30%	High Risk	Decline

Portfolio Metrics

metric	value_display
validation_applicants	61,503
average_probability_of_default	35.86%
actual_default_rate	8.07%
approval_rate	9.90%
manual_review_rate	38.02%
decline_rate	52.08%
total_expected_loss	5,678,645,380
loss_given_default_assumption	45.00%

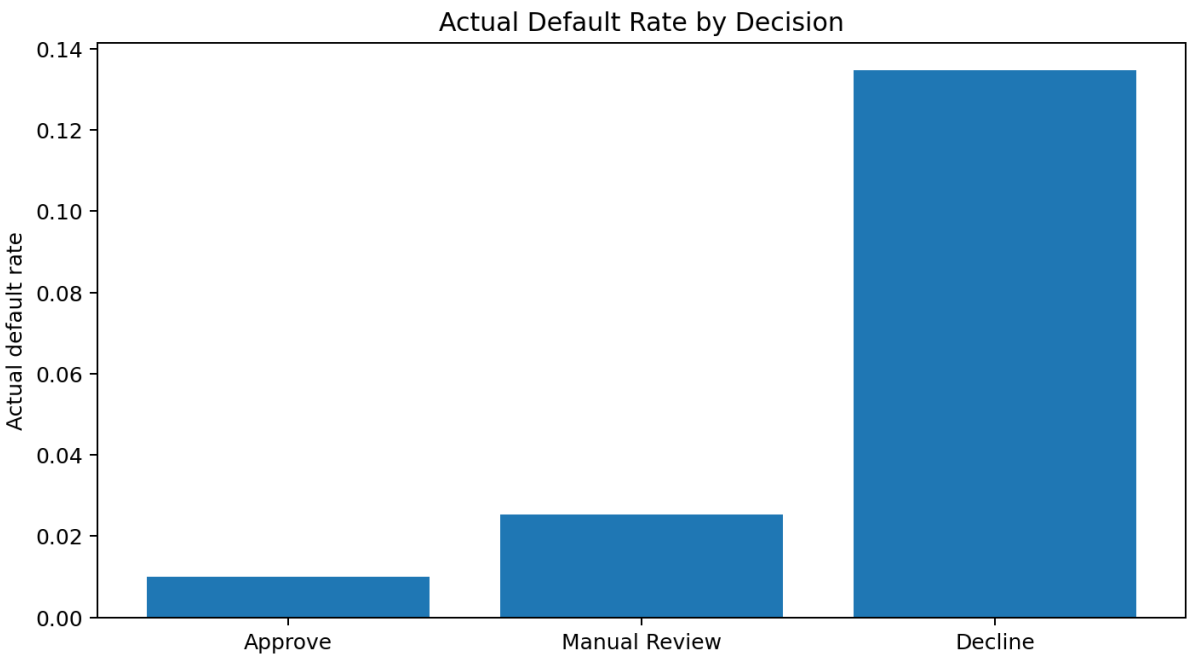
Note: exposure and expected-loss values are reported in dataset currency units, not pounds sterling.



Decision Distribution and Performance

recommended_decision	risk_band	applicant_count	average_probability_of_default	applicant_share
Approve	Low Risk	6087	6.98%	9.90%
Manual Review	Medium Risk	23383	19.55%	38.02%
Decline	High Risk	32033	53.25%	52.08%

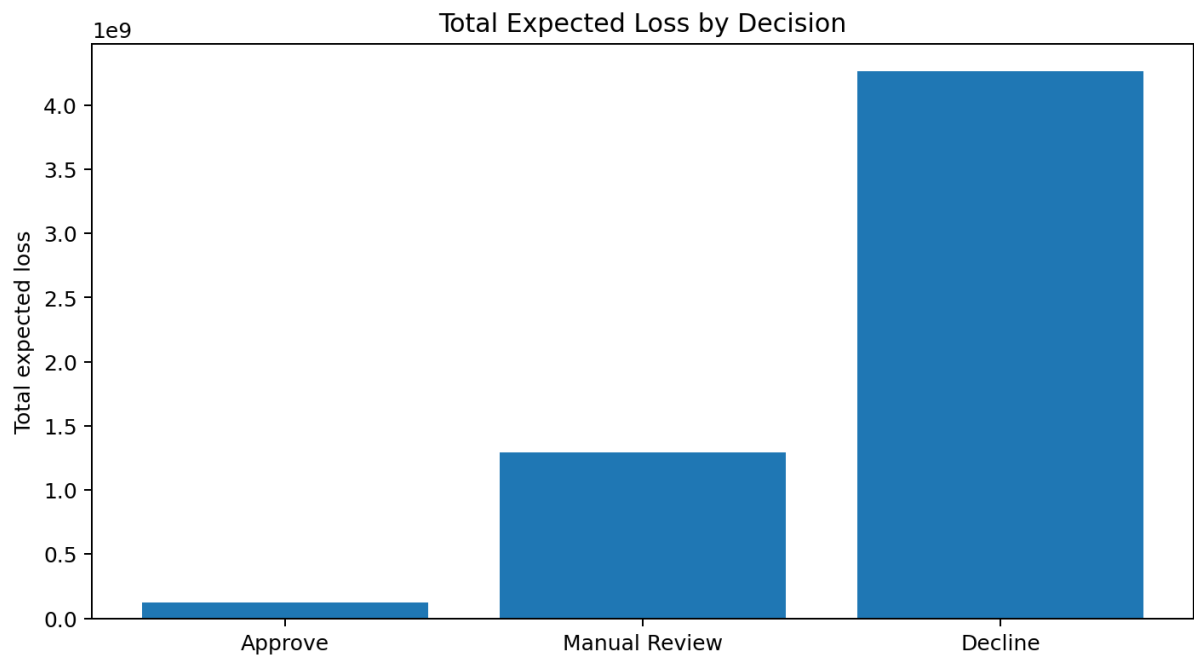
recommended_decision	risk_band	applicant_count	default_count	actual_default_rate	default_share
Approve	Low Risk	6087	61	1.00%	1.23%
Manual Review	Medium Risk	23383	592	2.53%	11.92%
Decline	High Risk	32033	4312	13.46%	86.85%



Expected Loss

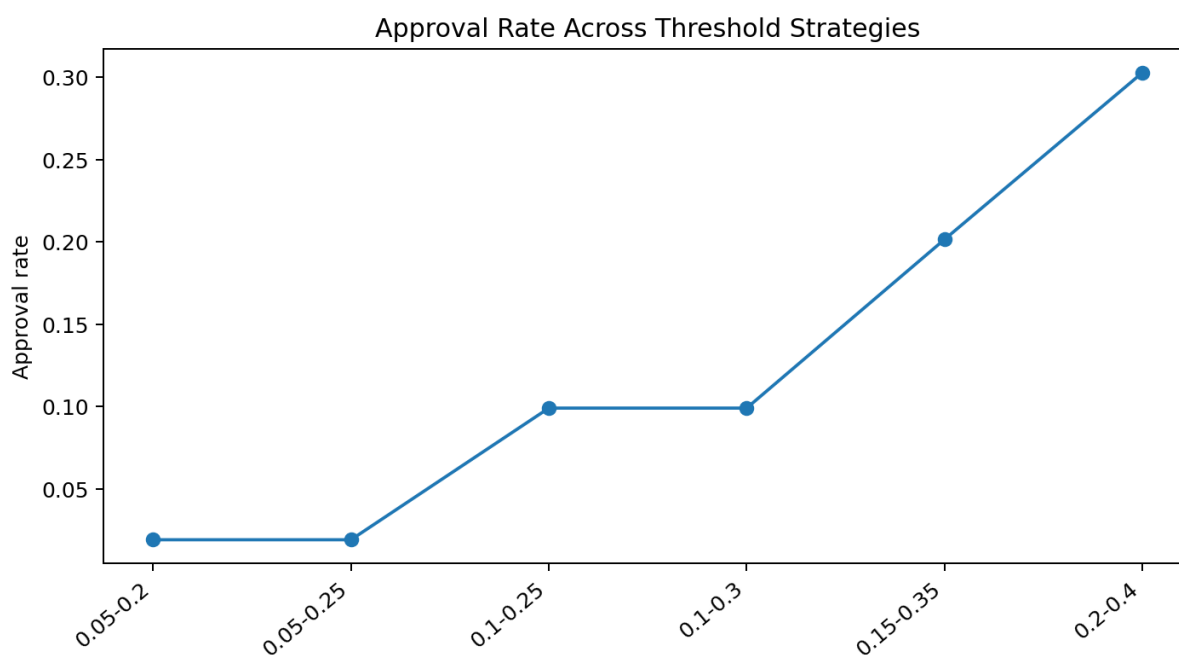
Expected loss is calculated as Probability of Default x Exposure at Default x Loss Given Default. The illustrative LGD assumption is 45%. Exposure and expected-loss values are expressed in the dataset currency units.

recommended_decision	risk_band	applicant_count	total_exposure	total_expected_loss	average_expected_loss
Approve	Low Risk	6087	3,851,499,154	120,558,941	19,806
Manual Review	Medium Risk	23383	14,762,851,448	1,296,049,524	55,427
Decline	High Risk	32033	18,150,729,543	4,262,036,915	133,051



Threshold Strategy Comparison

low_risk_thresh old	high_risk_thres hold	approval_rate	manual_review_r ate	decline_rate	approved_default_r ate	declined_default_rat e
0.05	0.20	1.90%	28.39%	69.72%	0.51%	10.91%
0.05	0.25	1.90%	37.58%	60.52%	0.51%	12.09%
0.10	0.25	9.90%	29.58%	60.52%	1.00%	12.09%
0.10	0.30	9.90%	38.02%	52.08%	1.00%	13.46%
0.15	0.35	20.15%	35.02%	44.83%	1.23%	14.84%
0.20	0.40	30.28%	31.41%	38.30%	1.54%	16.39%



Business Interpretation

The current thresholds are conservative. They create a small approved population with a low observed default rate and a larger decline segment containing most observed defaults. A lender could adjust thresholds depending on growth appetite, risk tolerance and manual review capacity.

Limitations

The analysis uses public data and illustrative assumptions. Production use would require affordability checks, compliance review, fairness testing, model monitoring, reject inference and governance controls.